

BA0-T1 concrete application trial

FR-3.4 Deliver RecognitionCapture - Revision 1

Repository baseline: bc0143eb8493ca91d3094eca9f8fbf9c7e6a3e58

Status: COMPLETED TRIAL / PROVISIONAL / BA0 NOT CLOSED

This trial applies the BA0 working hypothesis to one concrete facial-access fragment before any BAE taxonomy is designed. Trial identifiers are not metaclasses and no BAE type is accepted here.

1 What is being tested

The practical question is: which analytical knowledge must be made explicit around FR-3.4 so that the same system meaning can be reused across analysis consumers and controlled mutations?

The pinned source is:

```
04-specialized-requirement/03-example-facial-access/
DDTA_FACIAL_ACCESS_CAMERA_S1_STUDY_CORPUS_R2.tex
blob d54fdecc7d4968a88b74ea96343b37f38bb9b6d1
```

The source is an experimental DDTA study corpus. In this trial, “grounded” means directly supported by this pinned corpus, not production-system truth.

2 Source basis

- **D-3.3:** CameraSubsystem acquires evidence; a distinct local RecognitionProcessor performs recognition. The decision explicitly creates a responsibility separation and requires evidence to cross it.
- **FR-3.4:** CameraSubsystem delivers each remotely evaluated RecognitionCapture to RecognitionProcessor, correlates it with the correct RecognitionRequest, and must not report incomplete delivery as success.
- **D-3.4:** the project consumes local connectivity and does not own/manage the underlying transport infrastructure.
- **D-3.5:** current connectivity realization is wired Ethernet.
- **SR-3.4-C/I/P:** confidentiality, integrity and authorized-origin conditions apply to the FR-3.4 delivery/acceptance context.

The corpus also provides two mutations used here: M1 Ethernet to Wi-Fi, and M3 remote recognition to local recognition.

3 Trial notation

```
T-Rxx = provisional referent an analysis may need to distinguish
T-Sxx = provisional analytical statement about referents
T-Xxx = proposed analytical addition used as a falsification probe

G = grounded in selected source statement
D = derived analytical normalization, no new project commitment
A = analytical addition not asserted/entailed by the source
C = conflicting source support
```

Identity pressure means only that repeated analysis/change tracking appears to need the same referent to remain recognizable. It does not decide first-class BAE identity.

4 Pass 1 - provisional referents

ID	Label	Basis	Origin	Identity pressure
T-R01	CameraSubsystem	D-3.3, FR-3.4	G	YES
T-R02	RecognitionProcessor	D-3.3, FR-3.4	G	YES in V0; REVIEW under M3

ID	Label	Basis	Origin	Identity pressure
T-R03	RecognitionCapture	FR-3.3/3.4, SR-3.4-*	G	YES for concrete correlation; type-vs-instance OPEN
T-R04	RecognitionRequest	FR-3.3/3.4	G	YES for correlation
T-R05	consumed local connectivity/transport service	D-3.4	G	UNCERTAIN
T-R06	Ethernet realization	D-3.5	G	NO/UNCERTAIN; replaced by M1

No Actor, Asset, Channel, Interface, Flow, Boundary, State or InformationObject type is introduced by this list.

5 Pass 2 - minimum analytical statements

T-S01 – delivery structure (D). CameraSubsystem delivers RecognitionCapture to RecognitionProcessor and correlates it with RecognitionRequest. The graph tuple is derived normalization of FR-3.4, not a new project commitment.

T-S02 – completion semantics (G). Incomplete delivery is not successful completion.

T-S03 – separated responsibility placement (G). Acquisition and recognition are distinct in V0 under D-3.3.

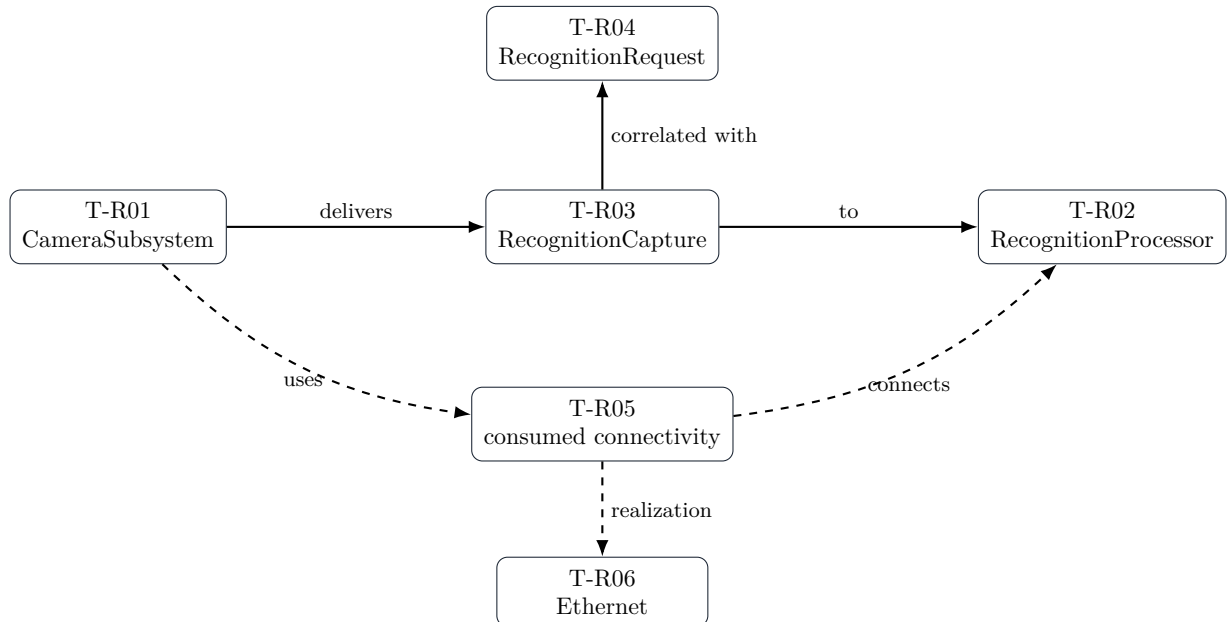
T-S04 – evidence crosses the separation (D). The RecognitionCapture must cross that separation in V0.

T-S05 – transport ownership context (G). The project consumes the transport and does not own/manage the underlying infrastructure in V0.

T-S06 – current realization (G). Connectivity realization is Ethernet.

T-S07/T-S08/T-S09 (G). Confidentiality, integrity and authorized-origin conditions constrain the FR-3.4 delivery/acceptance context.

6 What the trial representation looks like



The diagram is not authority

This rendering only visualizes the trial ledger. Source basis and analytical statements remain the semantic record.

7 Analytical-addition falsification probe

A tool could invent T-X01 `RecognitionIngressEndpoint`. The selected source does not name or require an independently meaningful ingress endpoint.

```
T-X01
origin      = A (candidate analytical addition)
accepted    = NO
reason      = baseline trial works without inventing it
```

The same applies to an invented first-class `NetworkChannel`. Transport semantics are real, but this trial does not force a channel entity with independent identity.

8 Reuse test

Two consumers reuse the same meaning without redefining it:

- **Consumer A - change-impact reasoning:** determines which statements survive Ethernet to Wi-Fi and which are revised.
- **Consumer B - SpecializedRequirement applicability reasoning:** determines which delivery/payload/request meaning is constrained by SR-3.4-C/I/P and whether those obligations remain applicable after mutation.

T1 result

A shared semantic contract is useful. T1 does not yet prove that DDTA must persist one long-lived materialized graph; a reproducibly rebuilt representation with stable identifiers remains a competing hypothesis.

9 Mutation M1 - Ethernet to Wi-Fi

Trial item	Result	Reason
T-R01/T-R02	RETAIN	same camera/processor separation
T-R03/T-R04	RETAIN	same capture/request meaning
T-R05	RETAIN	external ownership unchanged
T-R06	REPLACE	Ethernet becomes Wi-Fi
T-S01..T-S05	RETAIN	D-3.3, D-3.4 and FR-3.4 unchanged
T-S06	REVISE	realization changed
T-S07..T-S09	RETAIN after review	SRs retained in source regression

M1 shows why stable analytical identity is useful while simultaneously showing that Ethernet/Wi-Fi need not be first-class BAEs.

10 Mutation M3 - remote to local recognition

Trial item	Result	Reason
T-R01	RETAIN	camera remains relevant
T-R02	REVIEW / MAY RETIRE OR MERGE	separate responsibility meaning changes
T-R03/T-R04	REVIEW	local revised contract not inferred here
T-R05/T-R06	REVIEW	may be irrelevant to capture transfer but not necessarily to all project use
T-S01	RETIRE	old FR-3.4 retired/superseded
T-S02	RETIRE WITH OLD FR	scoped to old delivery
T-S03	REVISE/REPLACE	D-3.3 changed
T-S04	RETIRE	old crossing disappears
T-S07..T-S09	NOT APPLICABLE IN OLD FORM	source corpus says so

M3 shows that stable identity requires semantic continuity, not string equality. The name `RecognitionProcessor` alone is insufficient to assert that the same analytical referent survives.

11 Hypothesis results

Hypothesis	T1 result	Effect
Shared semantic contract is useful	SUPPORTED	keep in BA0
One persisted canonical graph is mandatory	NOT PROVEN / OPEN	weaken wording
Grounded / derived / addition distinction is useful	SUPPORTED WITH REFINEMENT	assertion-level origin pressure
Every useful responsibility needs first-class type	FALSIFIED	defer type decisions to BA1
Tool inference may become authority silently	REJECTED	T-X01 stays unaccepted

12 Revised BA0 candidate wording

REVISED WORKING CANDIDATE / NOT CLOSED

Base Analysis establishes an accepted, methodology-neutral analytical representation of the system knowledge needed by DDTA for a given governed baseline. The representation preserves shared analytical identity and the origin of analytical assertions, and can be reused or reproducibly rebuilt for multiple analysis consumers. It does not become a competing project source of truth, and no tool, view, method, or inferred structure may silently redefine its accepted meaning.

The key change is:

```
maintains one canonical persisted representation
-> NOT REQUIRED BY T1

establishes shared accepted analytical identity/meaning
-> SUPPORTED BY T1
```

13 What remains untested

- a real conflict between governed source claims;
- a case where an analytical addition is necessary rather than merely convenient;
- rename/refactoring identity rules;
- persisted representation versus reproducible rebuild;
- minimal BAE ontology, relation vocabulary, provenance implementation, views and threat overlays.

14 Next microstep

BA0-T2 - conflict + necessary analytical-addition stress test.

It will test one incompatible-source case, one genuinely necessary reviewed analytical addition, and a persisted-vs-rebuilt representation comparison. Only after T2 should BA0 be reconsidered for closure.

State after BA0-T1	
Chapters 2-4	CLOSED / FINAL
Documentation layer	CLOSED
W0	CLOSED
BA0-R systems-modeling prior art	CLOSED
BA0 working hypothesis R1	PRESERVED
BA0-T1 FR-3.4 application trial	COMPLETED / PROVISIONAL
BA0 responsibility/non-goals	REVISED WORKING CANDIDATE / NOT CLOSED
BA0-T2 conflict/addition trial	NEXT
BA1 minimal BAE ontology	NOT STARTED
BAE types accepted	NONE